

Interview Task: Building a Hybrid Query System (No Frameworks)

Objective:

We'd like you to design and implement a system that can answer user questions by intelligently querying both a structured database (SQL) and an unstructured text corpus (via vector search). The key challenge is to do this **without relying on existing LLM frameworks** like Langchain, LlamaIndex, etc. You will need to implement the core logic for query understanding, decomposition, data retrieval, and answer synthesis.

System Requirements:

1. **Input:** The system should accept a natural language question from the user.
2. **Outputs:** This final step should synthesize the information and generate a coherent, natural language answer to the user's original question.

Implementation Notes:

- **No Frameworks:** You must not use libraries like Langchain, Haystack, LlamaIndex, or any other comprehensive RAG/LLM agent frameworks. Standard libraries for database interaction (e.g., `sqlite3`), numerical operations (e.g., `numpy` for vector math if needed), and basic text processing are acceptable.
- **Frontend:** A command-line interface (CLI) for interaction is perfectly acceptable.
- **Focus:** The emphasis is on your understanding of the components of such a system, the logic flow, and how you would approach building it from more fundamental pieces.

Demonstration & Discussion:

During our meeting, we'd like you to:

1. **Demonstrate** your system with a few example questions.
2. **Talk us through your implementation:** Explain your design choices, the logic for query decomposition, data retrieval, result combination, and answer synthesis.
3. **Discuss the challenges** you encountered while building this system from scratch, particularly in the absence of high-level frameworks.